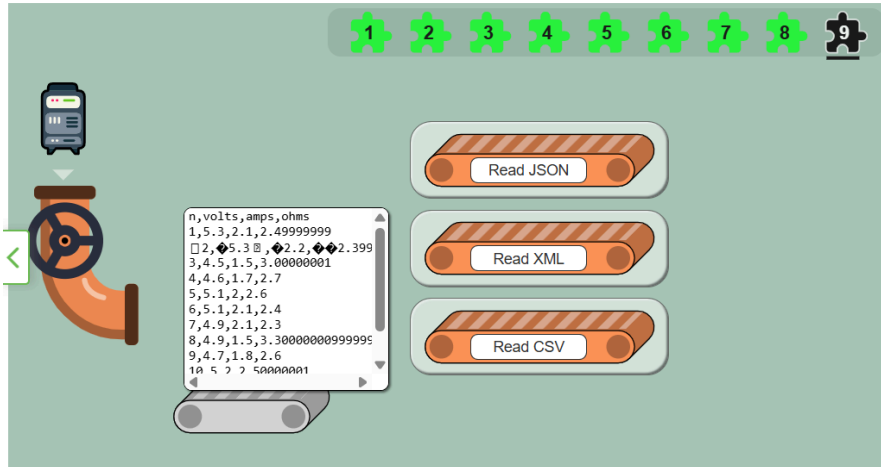
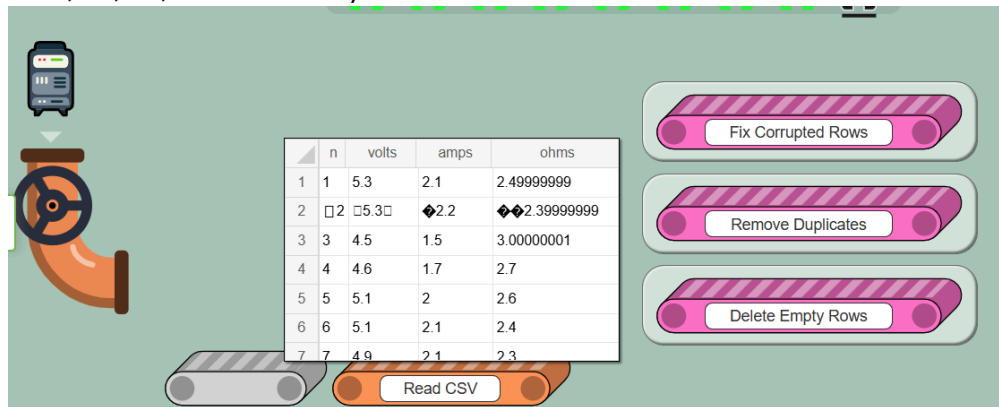


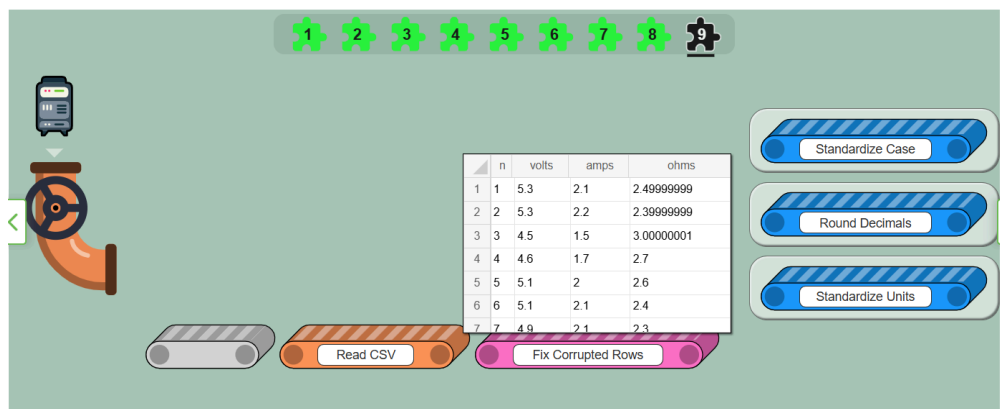
Hands-On Activity 2	
Course Code: CPE-031	Program: BsCpE
Course Title: Visualizations and Data Analysis	Date Performed: August 1, 2026
Section: CPE21S1	Date Submitted: August 1, 2026
Name: Carl Andrei D. Torres	Instructor: Engr. Jimlord Quejado
1. Discussion	
The Big Data processing workflow works like a continuous conveyor belt. It transforms raw input into high-quality, actionable data. The journey starts with Data Ingestion, where the external plain-text or structured files, such as CSVs, JSONs, and XML, are ingested and parsed into usable data structures. This step lays the foundation for all further data handling procedures. After ingestion, the data enters the Data Cleaning and Remediation Phase. In this phase, initial quality control operations which are remove corrupted symbols, fix structural errors, and deal with the missing values on each datapoints. This ensures that all data fields meet expected formats and prevents failures or errors in later processing. Once the data's structure is secure, it moves to Data Normalization and Standardization Phase. Here, precision errors, like floating-point rounding issues, are addressed to ensure logical errors are being corrected and formatting the modified values correctly across all variables. Finally, during Storage Phase, the fully cleaned and validated dataset reaches the end of the processing pipeline. In this last stage, the data is written and saved directly into the target database. This guarantees long-term data integrity and makes the dataset consistently available.	
2. Materials and Equipment	
Cisco NetAcad and Computer	
3. Procedure	
1. Read if the file reads in JSON, XML, or CSV	



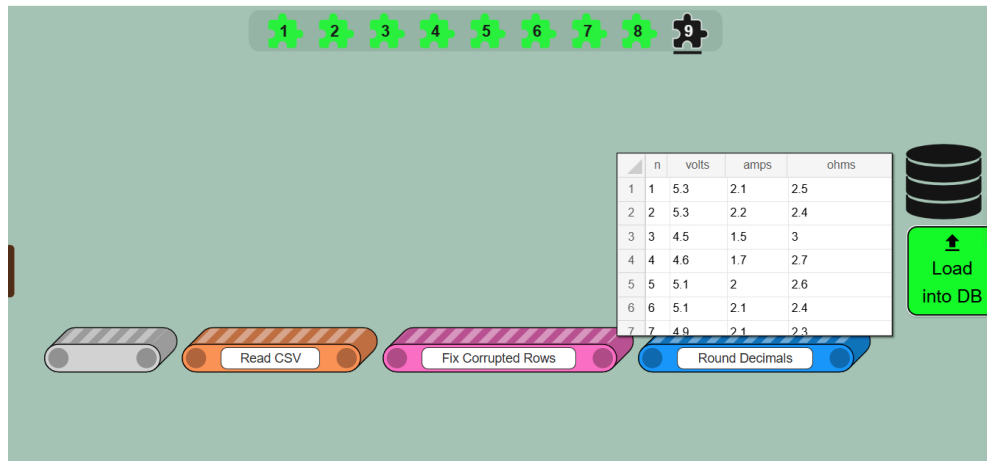
2. Read the file now and check if it needs to fix corrupted rows, remove duplicates, or delete empty rows (Corrupted Rows in Puzzle 9 which clean up the rows containing corrupted encoding characters (e.g., removing broken symbols like `` in row 2 to restore numeric values like 2, 5.3, 2.2, and 2.39999999).



3. Read the file now and check if it needs to standardized case, round decimals, or standardize units in Puzzle 9 (Round Decimals in Puzzle 9 because you need standardize the precision of the float numbers (e.g., 2.49999999 -> 2.5 and 3.00000001 -> 3)).



4. Load it in the database



4. Output

The screenshot displays a web browser with two tabs. The left tab shows a Cisco Networking Academy course page for 'CPE 031-CPE21S1 - 1 - 2627'. The course progress is 97%, and the module 'Data Pipeline Game' is completed. The right tab shows a user profile for 'CARLANDREI TORRES' with a profile picture, name, pronunciation, contact information, and biography. The browser's address bar shows the URL 'netacad.com/launch?id=cbe58042-1dd3-4006-8a78-ba17aa5f2842&tab=curriculum&view=c57186...'.

5. Conclusion/Analysis/Learnings

In Conclusion, I've learned that the **Big Data Text Files** need to be **verified and be modified** before it **get to the databases** which the process are like "**Conveyor Belts**" which the first process would start with **Data Ingestion**, where the first conveyor belt reads and converts raw input from external files, like CSVs, JSON's, and CSVs, into a structured format. The data then moves to **Data Cleaning and Remediation**, where **corrupted symbols, encoding errors, and invalid rows** are **removed to avoid system failures**. Next, it goes through **Data Normalization and Standardization** to **address precision issues**, such as **rounding floating-point numbers, ensuring consistent values across all variables**. Finally, the processed data reaches **Storage**, where the **clean and reliable dataset** is **loaded directly into the target database for safe querying and long-term use**.

6. Assessment Rubric

Lab Activity Rubric							
Criteria	Ratings						Pts
 SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	4 pts Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor Relies on classroom instruction only	1 pts Very Poor No initiative or interest in acquiring new knowledge	6 pts
 SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent Completes an assigned task independently and practices continuous improvement	5 pts Good Completes an assigned task without supervision or guidance	4 pts Satisfactory Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory Requires detailed or step-by-step instructions to complete a task	2 pts Poor Shows little interest to complete a task independently	1 pts Very Poor No interest to complete a task independently	6 pts
 SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory Apply the gathered information to formulate the problem	2 pts Poor Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor Gather information from a variety of sources	6 pts
 SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory Shows some creative ways to solve the problem	2 pts Poor Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor Ideas are copied or restated from the sources consulted	6 pts
Total Points: 24							



Please have the honor of pledge in the comment section or at the bottom of your activity document/exam or any form of file submission.

I affirm that I will not give or receive any unauthorized help on this activity/exam and that all work will be my own.